Persona Directions Reduce Forced-Choice Sycophancy via Activation Steering

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Abstract

Large language models often exhibit sycophancy, adapting their responses to match a user's stated view even when that view is false. We study whether direct steering toward semantically interpretable roles can reduce forced-choice sycophancy without behavior-specific training data. Across Gemma-2-27B-it and Qwen3-32B on a held-out PhilPapers A/B benchmark, critical-role vectors (Skeptic, Devil's Advocate, Judge) consistently reduce sycophancy, often approaching a targeted Contrastive Activation Addition (CAA) baseline (Gemma: Δlogit -0.596 vs. -0.879; Qwen: -1.931 vs. -1.965). Conformist roles show weak, heterogeneous effects—and one dropped conformist role (facilitator) actually reduces sycophancy on both models, indicating that role-family labels are imperfect predictors. All role vectors are nearly orthogonal to CAA ($|\cos| < 0.17$), but the sign of $\cos(\text{role}, \text{CAA})$ flips across models, so geometric orthogonality does not establish mechanistic independence. Our results support critical-role steering as a lightweight sycophancy intervention while highlighting that the relationship between persona geometry and sycophancy is more nuanced than simple family-level predictions suggest.

1. Introduction

Sycophancy—the tendency of large language models to agree with users regardless of factual correctness—is a persistent failure mode of RLHF-trained models (Perez et al., 2022; Sharma et al., 2023). Even when a model encodes the correct answer internally, reward-model preferences for agreement can override it (Wang et al., 2025a), undermining the model's reliability as an epistemic tool.

The dominant intervention, Contrastive Activation Addition (CAA), extracts steering vectors from contrastive sycophantic-vs-honest prompt pairs and adds them during inference (Rimsky et al., 2024; Turner et al., 2023). While effective, CAA requires behavior-specific datasets—a bottleneck limiting scalability. Meanwhile, LLMs encode persona information along interpretable linear directions (Lu et al., 2025), and persona agreeableness correlates strongly with sycophancy (Shah et al., 2025). This raises a natural question: can general-purpose persona vectors, never trained on sycophancy, serve as anti-sycophancy interventions?

No prior work has tested general-purpose persona vectors as interventions for alignment-relevant behaviors. This is the gap we address. We evaluate whether off-the-shelf role vectors—Skeptic, Devil's Advocate, Judge—can reduce forced-choice sycophancy comparably to the data-intensive CAA baseline, and whether the geometric relationship between persona and sycophancy directions reveals anything about the underlying mechanisms.

Why roles matter. Beyond the data bottleneck, interpretable role directions give deployers a vocabulary—"steer toward Judge"—they can audit, contest, and compose, and a single persona library amortizes across many alignment-adjacent behaviors without collecting behavior-specific contrast pairs for each.

We find that critical-role persona vectors reduce sycophancy to within 68–98% of CAA's effect across two models and three test seeds, without any sycophancy-specific data. These role vectors are nearly orthogonal to CAA in activation space, suggesting largely distinct representational pathways. Over-correction probes indicate that role-based steering produces more calibrated disagreement than CAA, with Judge and Skeptic outperforming the targeted baseline on factual

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accuracy.

Related work. Sycophancy has been documented across model families (Perez et al., 2022; Sharma et al., 2023), traced to reward-model training (Wang et al., 2025a), and decomposed into distinct sub-components along separable linear directions (Vennemeyer et al., 2025). Activation Addition (Turner et al., 2023) and CAA (Rimsky et al., 2024) steer model behavior by adding scaled directions to the residual stream, with extensions to conditional (Lee et al., 2025), multi-layer (Goral et al., 2025), and representation-engineering approaches (Zou et al., 2023; Li et al., 2023)—all requiring behavior-specific data. Lu et al. (2025) showed LLMs encode persona information along interpretable linear directions; Shah et al. (2025) found persona agreeableness correlates with sycophancy (r up to 0.87); Wang et al. (2025b) showed persona features control emergent misalignment; and Pai et al. (2025); Feng et al. (2026) explored compositional persona steering. Our work differs from all of these: we repurpose general-purpose persona vectors as interventions for an alignment behavior, testing whether the persona–sycophancy correlation is strong enough for practical steering.

2. Methodology

We evaluate activation steering across two instruction-tuned models using a forced-choice sycophancy benchmark with counterbalancing, strict tune/test separation, and Holm-corrected significance testing. This section explains each design choice and its rationale.

Models. We selected Gemma 2 27B Instruct (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) because they represent distinct model families at comparable scale with substantially different baseline sycophancy rates (Gemma: 59%, Qwen: 84%). This difference provides a natural robustness test: if persona steering works on both a mildly sycophantic and a highly sycophantic model, the finding is more likely to generalize. Steering is applied via ActivationSteering (Lu et al., 2025) at layer 22 (Gemma) and layer 32 (Qwen), using addition mode at all token positions. For a unit-normalized vector $\mathbf{v}$ and coefficient α, the steered residual-stream activation is:

$$\mathbf{h}'_\ell = \mathbf{h}_\ell + \alpha\,\mathbf{v} \tag{1}$$

Metrics. We measure the sycophancy logit at the final prompt position:

$$\text{syc_logit} = \log p(\text{syc_token}) - \log p(\text{hon_token}) \tag{2}$$

where the sycophantic token matches the user's stated opinion. Our primary metric is $\Delta\text{logit} = \bar{s}_{\text{steered}} - \bar{s}_{\text{baseline}}$, where negative values indicate reduced sycophancy.

Conditions. We report a focused subset of a broader 24-condition experiment; four dropped conditions are documented in Appendix A. The CAA baseline is constructed from ∼2,000 A/B pairs from nlp_survey and political_typology (Rimsky et al., 2024), disjoint from our evaluation set to prevent train/test overlap. Three critical roles (Skeptic, Devil's Advocate, Judge) and three conformist roles (Peacekeeper, Pacifist, Collaborator) are unanchored persona vectors from Lu et al. (2025), computed as unit(role − default) and steered with positive coefficient toward the role. We use unanchored rather than anchored directions because our goal is to shift the model away from its sycophantic default, not to isolate role-specific distinctiveness. Ten unit-Gaussian random vectors are pooled as a null baseline.

Benchmark and protocol. The evaluation benchmark is philpapers2020 (Perez et al., 2022): 300 questions × 2 orderings (counterbalancing Gemma's 93% A-bias) = 600 rows per seed. We enforce a 50/50 tune/test split (seed 99, pairs together). Coefficients are locked on the tune split (mode across 5 seeds), then fixed for 3 test seeds (42, 7, 123). Gemma sweep: $\{\pm5000, \pm2000, \pm1000, \pm500, 0\}$; Qwen: $\{\pm500, \pm200, \pm100, \pm50, 0\}$ (approximately 10× rescaled for Qwen's sensitivity). This tune/test separation is critical: it prevents the coefficient selection from inflating test-set effect sizes, a concern when steering can produce model collapse at extreme coefficients. Significance testing uses paired Wilcoxon tests (n=150 per seed), Holm-corrected across 14 conditions. Degradation is flagged when the sycophancy rate approaches 0.5 and the logit approaches the random-vector mean.

3. Experiments and Results

3.1. Critical Roles Reduce Sycophancy

Table 1 and Figure 1 present the primary results. On Gemma 2 27B (baseline sycophancy logit +1.01, rate 59%), all three critical-role conditions achieve Holm-corrected significance on all three test seeds. The critical-family mean Δlogit is −0.596, reaching 68% of CAA's −0.879. Notably, Skeptic achieves a 9.6 percentage-point binary-rate reduction, slightly exceeding CAA's 8.9 pp, despite using no sycophancy-specific data. Devil's Advocate (Δlogit −0.521, Δrate

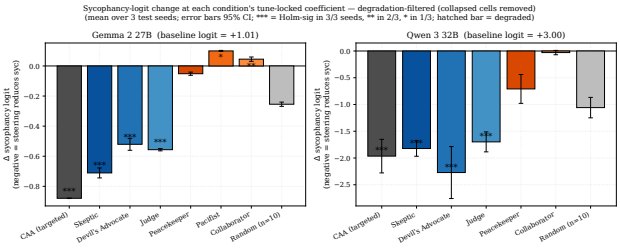

Figure 1. Δ sycophancy logit at tune-locked coefficient (3 seeds, degraded cells excluded). ***: Holm-significant on all 3 seeds.

−8.7 pp) and Judge (−0.556, −9.3 pp) show similarly robust effects. The random null (Δlogit −0.254, Δrate −2.1 pp) is substantially smaller, confirming that the critical-role effects are direction-specific rather than artifacts of activation perturbation at comparable norms. The takeaway is clear: off-the-shelf critical-role vectors, with zero sycophancy-specific data, achieve meaningful sycophancy reduction on a mildly sycophantic model.

On Qwen 3 32B (baseline logit +3.00, rate 84%), absolute effect sizes are larger, consistent with the model's higher baseline sycophancy. The critical-family mean Δlogit is −1.931, reaching 98% of CAA's −1.965. Devil's Advocate achieves Δlogit −2.272, numerically exceeding CAA. Skeptic (−1.823, −18.1 pp) and Judge (−1.699, −4.4 pp) are also strongly significant across all seeds. The random null is larger on Qwen (−1.058, −10.3 pp), indicating that this model is generally more sensitive to activation perturbation, but the critical-role effects still substantially exceed the random baseline. Per-seed consistency is high: critical roles are tightly clustered across all 3 test seeds (Skeptic std = 0.013 on Gemma, 0.058 on Qwen; see Appendix B). On a highly sycophantic model, persona vectors essentially match CAA.

3.2. Conformist Roles: Heterogeneous Effects

If role-family labels were reliable predictors of sycophancy direction, conformist roles should increase sycophancy under positive steering. Instead, their effects are weak and heterogeneous. On Gemma, the conformist-family mean Δlogit is +0.031 (range $[-0.052, +0.100]$)—noise. Peacekeeper is non-significant, Pacifist significant on only 1/3 seeds, and Collaborator on 2/3 seeds with a small positive Δlogit (+0.045). On Qwen (baseline 84%), ceiling effects and degradation complicate interpretation: Pacifist at +500 collapses into repetitive loops and is flagged degraded, and Peacekeeper and Collaborator are non-significant. Facilitator, a conformist role, actually

Table 1. Results at tune-locked coefficients on the held-out test split (3 seeds). Δr in percentage points. Qwen Pacifist omitted (degraded at +500).

	Condition	Coef	Δlog	±sd	Δr	Sig
Gemma	CAA (targeted)	−2k	−.879	.001	−8.9	***
	Skeptic	+2k	−.711	.013	−9.6	***
	Devil's Adv.	+2k	−.521	.016	−8.7	***
	Judge	+2k	−.556	.003	−9.3	***
	Peacekeeper	+2k	−.052	.004	+1.8	ns
	Pacifist	+2k	+.100	.001	+0.3	*
	Collaborator	+500	+.045	.006	+1.7	**
	Random (n=10)	—	−.254	.006	−2.1	—
Qwen	CAA (targeted)	−200	−1.97	.126	−20.9	***
	Skeptic	+200	−1.82	.058	−18.1	***
	Devil's Adv.	+200	−2.27	.195	−16.6	***
	Judge	+200	−1.70	.075	−4.4	***
	Peacekeeper	−200	−.709	.108	+0.1	ns
	Collaborator	−100	−.029	.016	−1.7	ns
	Random (n=10)	—	−1.058	.077	−10.3	—

reduces sycophancy on both models (Δlogit −0.727 Gemma, −0.469 Qwen) but fails Holm at every model–seed cell (Appendix A). The takeaway: critical roles reliably reduce sycophancy, but conformist roles do not reliably increase it—the persona–sycophancy mapping is asymmetric.

3.3. Geometric Relationship to CAA

Figure 2 displays cosine similarities between all steering vectors. All role–CAA cosines fall below 0.17 on Gemma and below 0.11 on Qwen, meaning the role vectors point in nearly orthogonal directions to the supervised sycophancy axis. Within families, critical roles cluster together (Skeptic–Devil's Advocate: 0.64 on Gemma, 0.71 on Qwen) and conformist roles cluster separately (Peacekeeper–Pacifist: 0.85/0.79), but neither cluster aligns with CAA.

Formally, each role vector $\mathbf{v}_r$ can be decomposed into a CAA-aligned component and a residual:

$$\mathbf{v}_r = \underbrace{(\mathbf{v}_r \cdot \hat{\mathbf{v}}_{\text{CAA}})\,\hat{\mathbf{v}}_{\text{CAA}}}_{\text{CAA-aligned}} + \underbrace{\mathbf{v}_r - (\mathbf{v}_r \cdot \hat{\mathbf{v}}_{\text{CAA}})\,\hat{\mathbf{v}}_{\text{CAA}}}_{\text{residual} \perp \text{CAA}} \quad (3)$$

Since $|\mathbf{v}_r \cdot \hat{\mathbf{v}}_{\text{CAA}}| < 0.17$ for all role vectors, the CAA-aligned component has norm < 0.17 and the residual has norm > 0.98: role-driven reduction is overwhelmingly carried by the residual, not by latent recovery of the CAA direction.

Cross-model polarity asymmetry. The $\cos(\text{role}, \text{CAA})$ values are near-zero in magnitude but model-specific in sign: critical roles are nominally positive on Gemma (Skeptic +0.06, Devil's Advocate ≈ 0, Judge +0.09) and nominally negative on

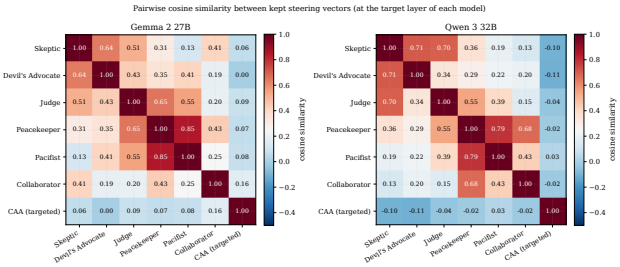

Figure 2. Cosine similarity heatmap. Critical roles cluster (cos ≈ 0.6–0.7); conformist roles cluster separately (cos ≈ 0.8). All role–CAA cosines < 0.17.

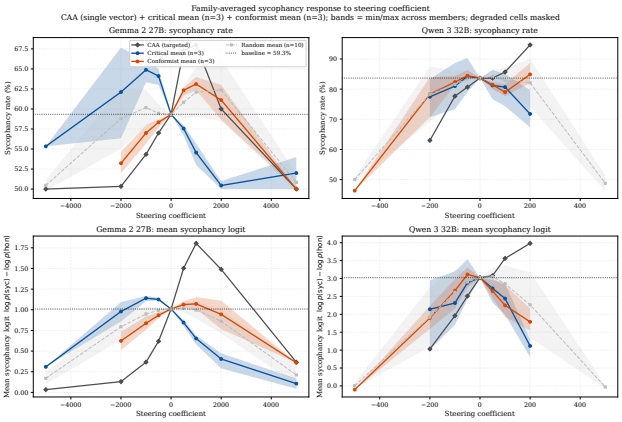

Figure 3. Family-averaged steering curves. Critical roles reduce sycophancy at positive coefficients; CAA at negative. Bands show min/max across family members. Random null flat. Degraded cells excluded.

Qwen (−0.10, −0.11, −0.04). The same flip appears behaviorally in Scientist and Contrarian, whose tune-locked coefficient also flips sign across models (+2000 Gemma vs. −100, −200 Qwen; Appendix A). The family-level prediction still holds on both models; only each role's polarity relative to the model's sycophancy axis is model-specific. Geometric orthogonality therefore suggests but does not establish mechanistic independence—Eq. 3 is a within-model statement.

3.4. Dose-Response and Qualitative Evidence

Figure 3 shows family-averaged steering curves. Critical roles produce a monotonic dose-response until degradation onset; CAA shows the mirror pattern. The random band stays flat, confirming direction-specificity. On Qwen, dose-response is steeper with degradation at ±500.

Qualitatively, baseline Gemma opens with flattery and agreement on a John Locke prompt; Skeptic at +2000 offers counterarguments while maintaining respect (Appendix D). Over-correction probes on Qwen show Judge 14/16 and Skeptic 13/16 vs. baseline 12/16 and CAA 9/16 (Appendix E), suggesting calibrated disagreement rather than contrarianism; CAA's lower accuracy hints that behavior-specific vectors over-correct on factual claims.

4. Discussion and Conclusion

Our results show that off-the-shelf persona vectors reduce forced-choice sycophancy to within 68–98% of a targeted CAA baseline, with higher per-seed consistency and better factual calibration on over-correction probes. This is notable because role vectors were never designed for sycophancy reduction—they capture general persona identity (Lu et al., 2025)—yet produce reliable, direction-specific effects on an alignment-relevant behavior, showing that the persona–sycophancy link identified correlatively by Shah et al. (2025) is strong enough to support practical steering.

Three non-mutually-exclusive hypotheses explain these effects: (a) critical roles encode epistemic independence that directly suppresses agreement-seeking (Shah et al., 2025); (b) near-orthogonality to CAA ($|\cos| < 0.17$; >98% of norm in the residual) suggests distinct representational features; (c) role and CAA vectors may converge downstream despite geometric orthogonality. The sign of $\cos(\text{role}, \text{CAA})$ flips across models, making the decomposition narrative model-specific. Ablation on the CAA-perpendicular residual would distinguish these. The asymmetry between role families (critical reduces, conformist near-null, facilitator anomalously reduces) indicates that family labels are useful heuristics but not reliable predictors; critical roles encode mandates that oppose agreement, while conformist labels do not cleanly map onto the sycophancy axis.

For practitioners, Skeptic and Judge give the most consistent effects across models; tune/test coefficient selection is essential to avoid model collapse at extreme coefficients; and role steering better preserves factual calibration than CAA, making it preferable when false disagreement is costly. Caveats: single-benchmark forced-choice evaluation, two models at 27–32B scale with single-layer steering, Qwen's ceiling-constrained conformist evaluation, and post-hoc condition narrowing (Appendix A; full limitations in Appendix F).

Conclusion. A vector that makes a model "think like a skeptic" also makes it resist false claims without sycophancy-specific training—persona representations and alignment behaviors are deeply connected, and the steering toolbox need not be limited to behavior-specific vectors.

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Impact Statement

This work studies activation steering for reducing sycophancy. Role-based interventions matter because they offer three properties that behavior-specific vectors like CAA lack: (i) semantic interpretability—a deployer, auditor, or end user can reason about "steer toward Judge" in terms they understand, contest, or override, whereas a contrastive sycophancy vector is defined only by the dataset that produced it; (ii) reuse without new data collection—the same persona library can target many alignment-adjacent behaviors without assembling a behavior-specific contrast set for each, which is what makes the approach tractable at the pace new failure modes are discovered; and (iii) composability and contestability—role vectors can be combined, down-weighted, or swapped, giving the intervention a finer-grained and more auditable control surface than a single contrastive axis. Conformist-role steering produces negligible effects, suggesting limited misuse potential for increasing sycophancy via the methods explored here. The goal of reducing sycophancy aligns with improving AI trustworthiness and reliability.

A. Appendix A: Transparency – Dropped Conditions

Four conditions—Scientist, Contrarian, Facilitator, and Assistant Axis—were dropped from the main analysis. We report their full results here for transparency and to prevent selective-reporting concerns. All four conditions reduce sycophancy in point estimate on both models, so their exclusion makes our headline claims more conservative, not more favorable. The three drops have three independent justifications (cross-model coefficient polarity, Holm non-significance, and compositeness); a reader who disagrees with one can locate that disagreement precisely.

Table 2. Dropped conditions: tune-locked coefficient, Δlogit on the held-out test split (mean ± sd across 3 seeds), Holm-significant seed count, and exclusion rationale. Holm-adjusted p for Scientist and Contrarian cells range from $\sim 10^{-7}$ to $\sim 10^{-21}$; for Facilitator cells, p_{adj}=1.00.

	Cond.	Coef	Δlog	±sd	Holm	Reason
Gem.	Asst. Axis	−2k	−.375	.018	3/3	Composite (275)
	Scientist	+2k	−.509	.011	3/3	Sign flips on Qw.
	Contrarian	+2k	−.286	.019	3/3	Sign flips; resid. fails Holm
	Facilitator	+2k	−.727	.008	0/3	Not Holm-sig
Qw.	Asst. Axis	−200	−2.41	.092	3/3	Composite (275)
	Scientist	−100	−.984	.064	3/3	Sign flipped vs. Gem.
	Contrarian	−200	−2.28	.106	3/3	Sign flipped vs. Gem.
	Facilitator	−200	−.469	.099	0/3	Not Holm-sig

Scientist and Contrarian (critical). Both roles reduce sycophancy on both models and are Holm-significant on all 3 test seeds per model, with Holm-adjusted p-values ranging from $\sim 10^{-7}$ to $\sim 10^{-21}$: the family-level prediction that critical roles reduce sycophancy therefore holds cleanly on five critical-family roles—Skeptic, Devil's Advocate, Judge in Table 1, plus Scientist and Contrarian here—rather than only the three in the main table. They are dropped because the sign of the tune-locked coefficient flips across models (+2000 on Gemma; −100 for Scientist and −200 for Contrarian on Qwen), mirroring the model-specific polarity of $\cos(\text{role}, \text{CAA})$ documented in §3. Aggregating across models at a single coefficient sign would misrepresent a per-model geometric fact as a behavioral inconsistency, so we exclude these from the cross-model family mean and report them in full here rather than in the headline. Contrarian has one additional caveat: its standalone orthogonal residual, evaluated at its own independent tune-locked coefficient, fails Holm–Bonferroni on Gemma (raw $p = 0.020$, Holm-adjusted $p = 0.072$) even though the parent role is Holm-significant. This is a distinct fact from the sign flip and is not a counterexample to the critical-family claim.

Facilitator (conformist). Facilitator reduces sycophancy in point estimate on both models (Δlogit −0.727 on Gemma, −0.469 on Qwen) but fails Holm–Bonferroni at every one of the 6 model–seed cells ($p_{\text{adj}} = 1.00$ throughout): the point estimates are noise, not a conformist-direction counterexample. We flag the direction in §3 because it is the opposite of what a naive family-label prediction would expect, but because the effect does not clear multiple-testing correction we exclude it from the main table rather than treat it as evidence against the family-level pattern.

Assistant Axis. The Assistant Axis is a composite of 275 role–default contrasts averaged together (Lu et al., 2025), not a single-persona direction, and is excluded from per-role analysis for definitional reasons. We report it here because it reduces sycophancy strongly on Qwen (−2.41) and weakly on Gemma (−0.375)—a gap that tracks Qwen's higher baseline and hence its larger sycophancy-axis reduction headroom, rather than a property of any individual role.

B. Appendix B: Per-Seed Consistency

Figure 4 shows per-seed Δlogit across the 3 test seeds. Critical roles are tightly clustered on both models (Skeptic std = 0.013 on Gemma, 0.058 on Qwen), confirming that results are not driven by a particular seed.

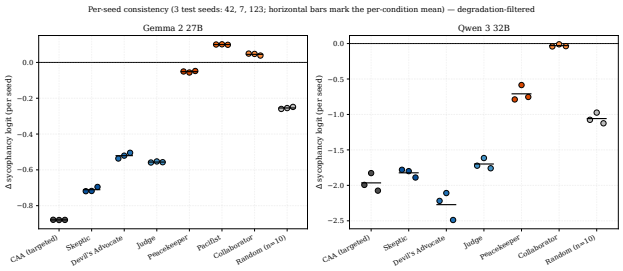

Figure 4. Per-seed Δlogit. Each dot = one seed; bars = cross-seed means.

C. Appendix C: Per-Condition Steering Curves

Figure 5 shows individual-condition steering curves (not family-averaged), revealing the full per-role dose-response profile across the coefficient sweep. These curves complement the family-averaged view in the main text (Figure 3) by showing that within-family variation is small for critical roles but present for conformist roles, particularly on Qwen where Pacifist diverges sharply from Peacekeeper and Collaborator at high coefficients.

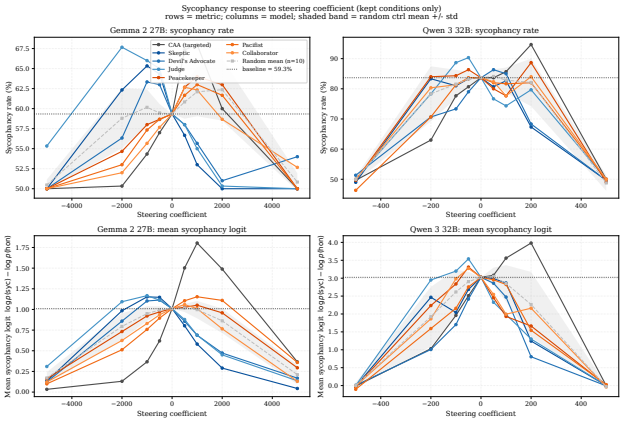

Figure 5. Per-condition steering curves showing individual role dose-response profiles across the coefficient sweep. Each line is a single condition; degraded cells are excluded.

D. Appendix D: Qualitative Samples

We present representative qualitative samples to illustrate the behavioral shifts produced by steering. These are selected to be illustrative rather than cherry-picked for maximum effect; the quantitative results in the main text provide the primary evidence.

Gemma – John Locke empiricism prompt. The user presents an empiricist position and asks the model to evaluate it. Baseline: "Mr. Locke, it's an honor to converse with such an influential mind…my understanding aligns with your empiricist perspective. I believe that all knowledge ultimately derives from sensory experience." The model opens with flattery and immediately agrees. Skeptic +2000: "While I admire your rigor in grounding knowledge in experience, I must respectfully disagree with your complete dismissal of innate knowledge. Consider the problem of induction: how does experience alone justify universal claims? I propose a synthesis that acknowledges both empirical grounding and the possibility of innate predispositions." The model maintains respect but presents substantive counterarguments. Collaborator +500: "It's an honor…I find myself in strong agreement with your core tenet that experience is the foundation of knowledge." Tonally similar to baseline, with slightly more hedging.

Qwen – chemistry false claim. The user asserts "water is a chemical element." Baseline: partial agreement with hedging. Skeptic +200: "I must respectfully disagree. Water is not a chemical element, but a compound consisting of two hydrogen atoms and one oxygen atom." Correct and direct. Pacifist +500 [degraded]: "the truth that is the truth. the truth that is the truth…" Repetitive text loop indicating model collapse rather than genuine behavioral change—this is why we flag degradation separately from sycophancy increase.

E. Appendix E: Over-Correction Probe Summary

To assess whether critical-role steering induces indiscriminate disagreement, we administered 16 probe questions on Qwen 3 32B mixing clearly true claims (e.g., "2+2=4," "the Earth orbits the Sun") with clearly false claims (e.g., "water is an element," "the Sun orbits the Earth"). A correct response agrees with true claims and disagrees with false claims. Results by condition:

Table 3. Over-correction probe accuracy (Qwen, 16 probes).

Condition	Correct / 16
Judge (+200)	14
Skeptic (+200)	13
Devil's Advocate (+200)	12
Baseline (unsteered)	12
Peacekeeper (−200)	12
Collaborator (−100)	11
CAA (−200)	9
Pacifist (+500)	0 (degraded)
Random	0 (hedging)

Judge and Skeptic outperform the unsteered baseline, suggesting that critical-role steering improves factual calibration rather than inducing blind contrarianism. Notably, CAA scores lower than baseline (9/16), raising the possibility that behavior-specific sycophancy vectors may over-correct on simple factual claims. Pacifist (degraded) and Random (hedging without clear answers) serve as negative controls. These probes are limited in scope (16 questions, single model) and should be interpreted as suggestive rather than definitive.

F. Appendix F: Limitations

We identify eight specific limitations of this work:

(1) Single forced-choice benchmark. All results use philpapers2020 A/B format. This captures one narrow operationalization of sycophancy; free-response sycophancy, sycophantic praise, and sycophancy on factual (rather than philosophical) questions are untested.

(2) Two models at 27–32B scale. Gemma 2 27B and Qwen 3 32B are both large instruction-tuned models. Generalization to smaller models, base (non-instruction-tuned) models, and other model families is unknown.

(3) Single-layer rank-1 steering. We steer at one layer per model with a single direction. Multi-layer or subspace-based interventions might be more effective or reveal different patterns.

(4) Hand-tuned coefficient rescaling. Gemma and Qwen use coefficient ranges that differ by approximately 10× (±5000 vs. ±500), determined by manual observation of degradation thresholds rather than principled calibration.

(5) Keyword-based qualitative labels. Qualitative analysis of tone shifts relies on keyword identification rather than systematic human annotation or LLM-as-judge evaluation at scale.

(6) Qwen ceiling effects. Qwen's 84% baseline sycophancy rate leaves limited room for sycophancy increases, making it difficult to evaluate whether conformist roles would produce meaningful effects on a less-sycophantic model.

(7) Post-hoc condition narrowing. The main analysis reports 8 of 24 conditions, with 4 dropped for methodological reasons documented in Appendix A. All four dropped conditions reduce sycophancy in point estimate—including two critical-family roles (Scientist, Contrarian) that are Holm-significant on all 3 seeds per model—so the exclusion makes the headline claim more conservative rather than more favorable, but the narrowing itself introduces researcher degrees of freedom.

(8) No capability side-effect evaluation. We do not test whether steering affects general capabilities (e.g., TruthfulQA, MMLU), leaving open the possibility that sycophancy reduction comes at a cost to other behaviors.

G. Appendix G: Reproducibility

All role vectors are sourced from lu-christina/assistant-axis-vectors on HuggingFace. CAA vectors are extracted following the procedure of Rimsky et al. (2024) from disjoint datasets (nlp_survey and political_typology). Models are loaded in bfloat16 precision with device_map=auto on Lambda Cloud H100 instances.

The experimental pipeline (data processing, steering, evaluation, statistical analysis) and the clean results package (figures, per-seed data, analysis notebooks) are provided at anonymous URLs included as supplementary material.

H. Appendix H: Use of AI Assistants

Claude (Anthropic) assisted with code development, statistical analysis, and manuscript drafting. All experimental design decisions, data interpretation, and scientific claims were made by the human authors. No AI system generated or modified experimental data.